Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Laser-Induced Synthesis of Nano-diamonds**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		**Chemicals**	
	Medium Category	Small Category	Medium Category	Small Category
	Ceramic Materials	Nano-Ceramic Composite Technologies	Fine Chemicals	Nano-Application Technologies
Concept of Proposed Technology	• Technology overview: The proposed technology is a laser-based processing platform for advanced carbon nanomaterials, enabling controlled synthesis of nanodiamond structures under ambient conditions. • The technology leverages pulsed laser–material interactions to induce localized extreme environments without relying on high-pressure or vacuum-based systems. • This approach offers high flexibility, scalability, and compatibility with existing industrial laser processing infrastructure.			
Rationale for Joint R&D Project	• Support is required to bridge the gap between laboratory-scale demonstrations and industrially relevant process validation for laser-induced carbon nanomaterial fabrication. • International collaboration with leading laser research institutes is essential to access advanced process optimization expertise and large-scale laser processing know-how.			

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	• International efforts in nanodiamond synthesis include electron-beam-based formation (Science, 2025) (1), molecular dynamics (MD) studies on diamond nucleation (Nature Comm., 2025) (2), and laser-assisted synthesis in liquid environments under ambient conditions (Nature, 2024) (3), reflecting active exploration of alternatives to conventional diamond fabrication. • Government support is necessary to enable Korean companies to participate in joint R&D, reduce technological and financial risks, and accelerate technology transfer toward commercialization.
Key Words	1. Laser-induced Synthesis 2. Nano diamonds 3. Nano materials 4. Localized Extreme Environments 5.

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (5)
R&D Objectives	Development of a laser processing framework for nanodiamond formation using pulsed laser–material interactions. ○ Systematic investigation of laser parameters (wavelength, fluence, pulse characteristics) and their influence on carbon nanostructure formation			
R&D Target Specifications	Establish a scalable laser-based processing technology for controlled synthesis of nanodiamond (99.5% purity) nanostructures under ambient conditions. ○ Demonstrate reproducible control of carbon phase transformation through laser parameter optimization.			
R&D Plan	• Define use cases, target specs, and baseline materials/precursors; consolidate physical models of pulsed laser–material interaction and diamond formation pathways (Year 1, TRL2→3). • Conduct systematic investigation of laser parameters (wavelength, fluence, pulse characteristics) and their influence on carbon nanostructure formation • Build a modular lab-scale laser platform with in-situ diagnostics; map process windows for nanodiamond vs. other carbon phases across different precursors (Year 1–2, TRL3→4). • Validate reproducible nanodiamond generation and properties (size, phase purity, yield, functional features like NV centers) in controlled lab conditions (end of Year 2, TRL4). • Transfer the process and framework to a relevant environment (pilot workstation), implement robust process control, and demonstrate application-oriented prototypes (coatings, particles) with industrial partners (Year 3, TRL5).			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Support is offered to bridge the gap between laboratory-scale demonstrations and industrially relevant process validation for laser-induced carbon nanomaterial fabrication. - International collaboration with leading laser research institutes is essential to access advanced process optimization expertise and large-scale laser processing know-how.
Expected R&R of Korean SME Partner	In Korea, laser processing technologies are actively studied in collaboration between universities and industry, particularly for advanced materials and functional carbon nanostructures. The proposed technology is currently being developed through joint research with Pusan National University, where academic–industrial cooperation is underway, including preparation of research publications and advanced material characterization to validate laser-induced carbon nanostructure formation.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	○Development of a laser-based process technology concept for controlled nanodiamond synthesis, building on existing laser processing capabilities and preliminary research outcomes. The approach aims to enable nanodiamond formation under ambient conditions without reliance on high-pressure equipment, hazardous gases, or toxic chemical precursors, offering potential environmental and safety advantages as well as prospects for cost-effective manufacturing using industrial laser systems. - The technology targets application-specific markets such as biosensing, quantum dot–related components, and high-performance sensor surfaces, where global demand is rapidly growing. By offering safer and lower-cost nanodiamond production, the project is expected to enhance the competitiveness of Korean companies in these emerging fields.

6. References

(1) Donadio, D., Berrens, M.L., Zhao, W. *et al.* Metastability and Ostwald step rule in the crystallisation of diamond and graphite from molten carbon. *Nat Commun* **16**, 6324 (2025). https://doi.org/10.1038/s41467-025-61674-5

(2) Costa, D.C., Fernandes, M., Moura, C. *et al.* Laser ablation in liquid-assisted synthesis of three types of nanoparticles for enhanced antibacterial applications. *Int. J. of Precis. Eng. and Manuf.-Green Tech.* **12**, 1699–1717 (2025). https://doi.org/10.1007/s40684-025-00711-6

(3) Fu J, Nakamuro T, Nakamura E. Rapid, low-temperature nanodiamond formation by electron-beam activation of adamantane C-H bonds. Science. 2025 Sep 4;389(6764):1024-1030. doi: 10.1126/science.adw2025. Epub 2025 Sep 4. PMID: 40906847.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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